CMP653 Database Management Systems

Term Project

Project Objective & Scope

The objective is to apply or improve state-of-the-art database management tools, algorithms, and architectures. Projects should ideally be a mix of the following three types:

- **Experimental Evaluation:** Rigorous testing of database models or algorithms on interesting, large-scale datasets.
- **Theoretical Research:** Proposing a model, algorithm, or database property and deriving rigorous results or proofs.
- **Scalable Implementation:** Developing high-performance implementations of algorithms for processing massive data or specialized database kernels.

Every project must contain a degree of **mathematical analysis** and **experimentation** on real or synthetic data.

Evaluation Criteria

Projects will be assessed based on the following professional standards:

- **Technical Quality:** Does the material make sense, and are the methods reasonable?
- **Innovation:** Are the proposed applications clever? Do you provide novel insights?
- **Significance:** Did you choose a "real" problem rather than a "toy" problem? Is the work likely to have an impact?
- **Presentation:** Clarity of the write-up and novelty of the work.

Deliverables

The project is divided into three main milestones to ensure continuous progress.

- Project proposal (20% of the project grade) **due: 16/3/2026**
- Project milestone report (20% of the project grade) **due: 6/4/2026**
- Final project report (60% of the project grade) **due: 12/5/2026**

1. Project Proposal (20% of Project Grade)

This consists of two related parts totaling 3–5 pages.

- **Part 1: Reaction Paper (2–3 pages):** You must select three or more research papers related to CMP653 topics (e.g., HTAP, Vector DBs, or Autonomous Tuning).
  - **Summary:** Detail the technical content and the connection between the papers.
  - **Critique:** Identify strengths, weaknesses, and unrealistic assumptions.
  - **Brainstorming:** Propose research questions, better models, or new algorithms based on your reading. This should be the most substantial part.
- **Part 2: Formal Proposal (1–2 pages):** Building on your brainstorming, define:
  - The specific problem and the data you will use.
  - The algorithms, techniques, or models you plan to develop.

    - Your evaluation plan: How will you measure success?

2. Project Milestone Report (20% of Project Grade)

Think of this as a draft of your final report with roughly **40% of the work completed**. It should be 3–5 pages and include:

- A thorough problem introduction and review of prior work.
- Data collection descriptions and initial findings/statistics.
- Necessary mathematical background and formal algorithm descriptions.
- An outline of the remaining tasks.

3. Final Project Report (60% of Project Grade)

The final output is an 8–10 pages research paper (maximum 12 pages). It will be graded according to this rubric:

- **Introduction/Motivation (15%):** Why does this problem matter?
- **Related Work (10%):** A summary of cited papers and how they relate to your work.
- **Methodology (30%):** A clear, detailed description of your primary technical contribution.
- **Results and Findings (35%):** Conclusive experiments that evaluate your solution. Theoretical projects must include proofs or runtime analysis here.
- **Style and Writing (10%):** Organization, grammar, and neatness.